

Matrix inequalities and norms

Sharp additive triangle constant for an odd number of contractions

Difficulty: challenging **Importance:** interesting to specialist

Status: [Lean verified](#)

Rating rationale: Sharpness for every odd summand count needs a new extremal construction or obstruction; the payoff is concentrated in operator triangle inequalities.

Resolution — 2026-09-11

Affirmative result by Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. **Independent proof review:** **PASS**.

The sharp additive contraction constant is $c_k = k/4$ for every $k \geq 2$, including every odd summand count. Exact dimension-two extremizers match the universal upper bound.

The exact target is resolved. The original statement and source evidence are retained below; its former difficulty rating is historical.

Primary manuscript: [complete proof PDF](#), [standalone TeX](#), Theorem 1.1 and its proof; [authorship and scope](#). The [independent review](#) checks the full original argument and records its hash. The draft was AI-assisted; this is independent agent verification, not external human peer review or formal certification. [Submission record](#).

Lean proof and verification evidence — 2026-09-14

The complete original target is Lean verified. The complete original odd-summand conjecture is proved and, more strongly, the sharp additive contraction constant is $k/4$ for every $k \geq 2$. The formalization uses the actual complex Euclidean operator norm, positive square-root matrix modulus and positive-semidefinite order, with genuine infima of nonempty sets bounded below and exact dimension-two extremizers.

Immutable formal proof. The checked exports are `NLA.MI03.upper_bound`, `NLA.MI03.sharpness`, `NLA.MI03.sharp_constant`, `NLA.MI03.odd_sharp_constant`. Lean 4.33.1 uses pinned Mathlib `0df444a3` and LeanCert `621a43d7`; [complete pins and reproduction instructions](#) are retained. All four exported transitive axiom closures contain only `propext`, `Classical.choice` and `Quot.sound`.

Formalization: Sidney Holden, Center for Computational Biology, Flatiron Institute, Simons Foundation, with OpenAI Codex assistance. Matthew J. Colbrook retains mathematical authorship. Two independent statement reviews preceded implementation, and two independent final proof reviews passed. Kernel-only LeanCert and [fresh isolated Linux Comparator and kernel verification](#) passed, including rejection controls. [Original artifact](#), [exact checked inputs](#) and [independent operational audit](#) are retained. This is AI-assisted formal verification, not external human peer review or source-author endorsement.

Problem statement

For each integer $k \geq 2$, let c_k be the infimum of all $c \geq 0$ such that

$$\left| \sum_{j=1}^k A_j \right| \leq cI_n + \sum_{j=1}^k |A_j|$$

for every $n \geq 1$ and all $A_1, \dots, A_k \in \mathbb{C}^{n \times n}$ satisfying $\|A_j\|_2 \leq 1$. Here $|A| = (A^*A)^{1/2}$, $\|\cdot\|_2$ denotes the operator norm, and $X \leq Y$ means $Y - X$ is positive semidefinite.

Is $c_k = k/4$ for every odd integer $k \geq 3$?

Why it matters

The scalar absolute-value triangle inequality fails in matrix order. The optimal additive correction quantifies that failure for uniformly bounded summands, in a form usable in positive block-matrix estimates.

References

1. J.-C. Bourin and E.-Y. Lee, *Diagonal and off-diagonal blocks of positive definite partitioned matrices*, arXiv:2307.02034v3 (15 December 2023), Corollary 4.4 and the conjecture immediately following Remark 4.5. [Primary text](#).
2. E.-Y. Lee, *How to compare the absolute values of operator sums and the sums of absolute values?*, *Operators and Matrices* 6(3) (2012), 613–619; §2 supplies related triangle inequalities. [Primary paper](#), [DOI](#).

Status check — 2026-09-10

Corollary 4.4 establishes $c_k \leq k/4$ for every k and sharpness for even k ; the source separately conjectures sharpness for all odd $k > 1$. Its latest version remains v3. Searches included *Bourin Lee contractions odd k constant k/4*, *three contractions 3/4 sharp conjecture*, and *Bourin contractions sharp 2025 2026*. The 2026 symmetric-modulus papers concern different inequalities. No resolution of this additive odd-summand problem was located.

Audit update (2026-09-10): Rechecked the conjecture after Remark 4.5 in Bourin–Lee and searched for odd-summand sharpness results. The proved even-summand statement is outside the target, which already restricts to odd k . This is a bounded literature check, not a proof that no solution exists.